1 Curves

Definition 1. A map $\mathbf{f}:\mathbb{R}^n\to\mathbb{R}^m$ is **differentiable at $\mathbf{p}$** if there exist a *linear* map $D_{\mathbf{p}}\mathbf{f}:\mathbb{R}^n\to\mathbb{R}^m$ such that:

$$\lim_{\mathbf{h}\to 0}\frac{\|\mathbf{f}(\mathbf{p}+\mathbf{h})-\mathbf{f}(\mathbf{p})-D_{\mathbf{p}}\mathbf{f}(\mathbf{h})\|}{\|\mathbf{h}\|}=0$$

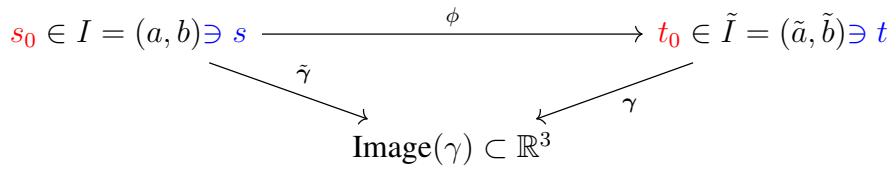
Note: $D_{\mathbf{p}}\mathbf{f}(\mathbf{v})=D_{\mathbf{p}}\mathbf{f}\mathbf{v}$, matrix multiplication always, since it is a linear map (at $\mathbf{p}$).

Examples: multivariable function $\mathbf{f}(x_1,x_2\cdots x_n)=(f_1(\mathbf{x}),f_2(\mathbf{x}),\cdots,f_m(\mathbf{x}))$, then $(D_{\mathbf{p}}\mathbf{f})_{ij}=\left.\dfrac{\partial f_i}{\partial x_j}\right|_{\mathbf{p}}$.

Theorem 1 (Chain Rule). $D_{\mathbf{p}}(\mathbf{f}\circ\mathbf{g})=D_{\mathbf{g}(\mathbf{p})}\mathbf{f}\circ D_{\mathbf{p}}\mathbf{g}$. *The useful identities are when we unit-speed reparametrize as $\widetilde{\boldsymbol{\gamma}}(s)=\boldsymbol{\gamma}(\phi(s))$ for any regular $\boldsymbol{\gamma}$ and $\widetilde{\boldsymbol{\gamma}}$ unit speed.*
Note that ϕ is monotone, i.e. either $\phi'(s)>0$ for all s or $\phi'(s)>0$ for all s.

- $\widetilde{\boldsymbol{\gamma}}'(s)=\boldsymbol{\gamma}'(\phi(s))\phi'(s)$, *for which we abbreviate as $\widetilde{\boldsymbol{\gamma}}'=\boldsymbol{\gamma}'\phi'$, take extra care on which what is inside the brackets. Taking modulus we have $1=\pm\|\boldsymbol{\gamma}'\|$. Hence we can deduce useful formulas like:*
- $\|\phi'\|^2=\dfrac{1}{\|\boldsymbol{\gamma}'\|^2}$, $\quad\widetilde{\boldsymbol{\gamma}}''=\boldsymbol{\gamma}''(\phi')^2+\boldsymbol{\gamma}'\phi''$, $\quad\phi''=\dfrac{-\boldsymbol{\gamma}'\cdot\boldsymbol{\gamma}''}{\|\boldsymbol{\gamma}'\|^4}$ *(from 2021 Exam 1.e)*

Commuting Reparametrizations:



Theorem 2. *For every regular curve $\boldsymbol{\gamma}$, if we take $\phi^{-1}(s)=\displaystyle\int_0^s\|\boldsymbol{\gamma}'(t)\|\,dt$, then $\widetilde{\boldsymbol{\gamma}}(s)=\boldsymbol{\gamma}(\phi(s))$ is always unit speed. Also, for any constant speed curve $\boldsymbol{\gamma}$, $\boldsymbol{\gamma}'\cdot\boldsymbol{\gamma}''=0$.*

Definition 2. The **curvature** at a point is $\kappa_{\boldsymbol{\gamma}}(t)=\kappa_{\widetilde{\boldsymbol{\gamma}}}(s)=\|\widetilde{\boldsymbol{\gamma}}''(s)\|=\dfrac{\|\boldsymbol{\gamma}''\times\boldsymbol{\gamma}'\|}{\|\boldsymbol{\gamma}'\|^3}$ in $\mathbb{R}^3$.

Definition 3 (Curves in $\mathbb{R}^2$)**.** Let $\boldsymbol{\gamma}:(a,b)\to\mathbb{R}^2$ be a unit-speed smooth curve. At $\boldsymbol{\gamma}(s)$, denote the (unit-length) **tangent vector** by $\mathbf{t}=\mathbf{t}(s)=\boldsymbol{\gamma}'(s)$. The **signed unit normal** is the vector obtained by rotating $\mathbf{t}$ by $\pi/2$ counterclockwise, i.e. $\mathbf{n}_s=J\mathbf{t}$ where $J=\begin{pmatrix}0&-1\\1&0\end{pmatrix}$. The **signed curvature** $\kappa_s=\kappa_s(s)$ of $\boldsymbol{\gamma}$ at $\boldsymbol{\gamma}(s)$ is defined to be $\kappa_s(s)\in\mathbb{R}$ so that $\boldsymbol{\gamma}''(s)=\kappa_s(s)\mathbf{n}_s(s)$.

Definition 4 (Curves in $\mathbb{R}^3$)**.** Let $\boldsymbol{\gamma}:(a,b)\to\mathbb{R}^3$ be a unit-speed smooth curve. The **tangent vector** is still $\mathbf{t}=\mathbf{t}(s)=\boldsymbol{\gamma}'(s)$. The **principal normal vector** of γ at $\gamma(s)$ is the vector $\mathbf{n}(s)=\dfrac{1}{\kappa(s)}\mathbf{t}'(s)$. The **binormal vector** of γ at $\gamma(s)$ is the vector $\mathbf{b}(s)=\mathbf{t}(s)\times\mathbf{n}(s)$. The **torsion** $\tau(s)$ is the value such that $\mathbf{b}'(s)=-\tau(s)\mathbf{n}(s)$.

Theorem 3 (Fernet-Serret)**.** *Let $\boldsymbol{\gamma}:(a,b)\to\mathbb{R}^3$ be a unit-speed smooth curve.* $\begin{pmatrix}\mathbf{t}'\\\mathbf{n}'\\\mathbf{b}'\end{pmatrix}=\begin{pmatrix}0&\kappa&0\\-\kappa&0&\tau\\0&-\tau&0\end{pmatrix}\begin{pmatrix}\mathbf{t}\\\mathbf{n}\\\mathbf{b}\end{pmatrix}$

Theorem 4. *If $\boldsymbol{\gamma}:(a,b)\to\mathbb{R}^3$ is a smooth regular curve For all $t\in(a,b)$, with non-zero curvature (so that $\mathbf{n}$ is defined.), then* $\mathbf{t}(t)=\pm\dfrac{\boldsymbol{\gamma}'(t)}{\|\boldsymbol{\gamma}'(t)\|}$, $\mathbf{b}(t)=\pm\dfrac{\boldsymbol{\gamma}'(t)\times\boldsymbol{\gamma}''(t)}{\|\boldsymbol{\gamma}'(t)\times\boldsymbol{\gamma}''(t)\|}$, $\tau(t)=\dfrac{(\boldsymbol{\gamma}'(t)\times\boldsymbol{\gamma}''(t))\cdot\boldsymbol{\gamma}'''(t)}{\|\boldsymbol{\gamma}'(t)\times\boldsymbol{\gamma}''(t)\|^2}$

2 Geometry

2.1 Spherical Geometry

Definition 5. Spherical Lines: Intersection with plane through origin, line segment are **NOT** uniquely determined. **Angle** is the angle of two plane **always defined to be less than 90**, distance is the shorter side $R\theta$. The distance is a metric, with triangle inequality proven by spherical first cosine law.

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be the position vectors of three points, respectively. Observe that $\mathbf{a} \cdot \mathbf{b} = R^2 \cos(c/R), \mathbf{a} \cdot \mathbf{c} = R^2 \cos(b/R)$, and $\mathbf{b} \cdot \mathbf{c} = R^2 \cos(a/R)$. A unit normal vector to the plane OBC is $\mathbf{n}_a$ where

$$\mathbf{c} \times \mathbf{b} = \|\mathbf{c}\|\|\mathbf{b}\| \sin(a/R)\mathbf{n}_a = R^2 \sin(a/R)\mathbf{n}_a.$$

The other two equations permutes from $\mathbf{c} \to \mathbf{b} \to \mathbf{a}$.

Theorem 5. $\sin\frac{a}{R}\sin\frac{b}{R}\cos\angle C = \cos\frac{c}{R} - \cos\frac{a}{R}\cos\frac{b}{R}, \frac{\sin\frac{a}{R}}{\sin\angle A} = \frac{\sin\frac{b}{R}}{\sin\angle B} = \frac{\sin\frac{c}{R}}{\sin\angle C}, \sin\angle A \sin\angle B \cos\frac{c}{R} = \cos\angle C + \cos\angle A\cos\angle B$

Theorem 6. *Area of a spherical triangle is $R^2(\angle A + \angle B + \angle C - \pi)$, isometries are orthogonal matrices (i.e. rotation of the sphere).*

2.2 Riemannian Metric

Definition 6. A **Riemannian Metric** on open $U \in \mathbb{R}^2$ is a smooth choice of inner product for all $\mathbf{p} \in U$. Ways to check is to write $\langle \mathbf{v}, \mathbf{w} \rangle_{\mathbf{p},U} = \mathbf{v}^T \left.\begin{pmatrix} E & F \\ F & G \end{pmatrix}\right|_{\mathbf{p}} \mathbf{w} = E\,du^2 + 2F\,du\,dv + G\,dv^2$, then check $E > 0, EG - F^2 > 0$ for all $\mathbf{p} \in U$. $E, F, G : U \to \mathbb{R}$ can be recovered by calculating $\langle \mathbf{e}_1, \mathbf{e}_1 \rangle, \langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \langle \mathbf{e}_2, \mathbf{e}_1 \rangle$, and $\langle \mathbf{e}_2, \mathbf{e}_2 \rangle$ respectively.

Definition 7. Length of a curve γ (which lives in U) with respect to a Riemannian metric is $\int_a^b \sqrt{\langle \gamma'(t), \gamma'(t) \rangle_{\gamma(t)}}\, dt$. The **Riemannian distance** between two points in U is the infimum of curve lengths. It is a metric.

Definition 8. The **area** of a region $V \subset U \subset \mathbb{R}^2$ under a Riemannian metric E, F, G is: $\iint_V \sqrt{EG - F^2}\, du\, dv$

Definition 9. Suppose $U, \widetilde{U} \subset \mathbb{R}^2$ are open sets with Riemannian metrics $\langle \cdot, \cdot \rangle_Q$ for $Q \in U$ and $\langle \cdot, \cdot \rangle_P^{\sim}$ for $P \in \widetilde{U}$. A **Riemannian isometry** $\mathbf{f} : \widetilde{U} \to U$ is a diffeomorphism so that for all $\mathbf{p} \in \widetilde{U}$ and for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$,

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{p}}^{\sim} = \langle D_{\mathbf{p}}\mathbf{f}(\mathbf{x}), D_{\mathbf{p}}\mathbf{f}(\mathbf{y}) \rangle_{\mathbf{f}(\mathbf{p})}.$$

Recall that a map $\mathbf{f} : \widetilde{V} \to V$ between two open subsets $\widetilde{V}, V$ of $\mathbb{R}^n$ is called a **diffeomorphism if** $\mathbf{f}$ is a bijection, and $\mathbf{f}$ and $\mathbf{f}^{-1}$ are smooth.

Theorem 7. *Riemannian isometries preserves length and area, this moreover implies that riemannian isometries are isometries considered over the metric space $\langle \tilde{U}, \tilde{\rho} \rangle \to \langle U, \rho \rangle$.*

2.3 Hyperbolic Geometry

Definition 10 (Upper Half Plane Model)**. Lines** are semicircles centered at x-axis or vertical lines. Isometries are finite composition of reflections and **inversion among lines.**, it can also be realized as composition of reflections **among vertical lines**, dilations and standard inversion $\mathcal{I}_{\mathbb{S}^2}$ (**among lines** means hyperbolic lines.).

The model follow the **hyperbolic metric** $ds^2 = \frac{du^2 + dv^2}{v^2}$. That is, $\langle \mathbf{v}, \mathbf{w} \rangle_{\mathbf{p}=(u,v)} = \frac{1}{v^2}(\mathbf{v} \cdot \mathbf{w})$

Definition 11. The **inversion** at a circle centered at $\mathbf{p}$ with radius r has the formula:

$$\mathcal{I}_{\mathbf{p},r}(\mathbf{x}) = \begin{cases} \mathbf{p} + \dfrac{r^2(\mathbf{x} - \mathbf{p})}{\|\mathbf{x} - \mathbf{p}\|^2} & \text{if } \mathbf{x} \in \mathbb{R}^2 \backslash \{\mathbf{p}\} \\ \infty & \text{if } \mathbf{x} = \mathbf{p}, \text{ and} \\ \mathbf{p} & \text{if } \mathbf{x} = \infty \end{cases}$$

Note that $\mathcal{I}_{\mathbf{p},r}\left[\mathcal{I}_{\mathbf{p},r}\right](\mathbf{x}) = \mathbf{x}$, just like reflection, so actually it is a bijection.

Theorem 8. *Suppose $\gamma : (a, b) \to H$ is a piecewise smooth curve from P to Q in H. Then the hyperbolic length $L(\gamma) \geq d_H(P, Q)$ with equality if and only if γ parametrises the hyperbolic line segment PQ.*

The theorem above ensure the definition of hyperbolic lines is the correct one.

Theorem 9. *A hyperbolic triangle has area $\pi - \angle A - \angle B - \angle C$*

3 Surfaces

Definition 12. A set $S \subset \mathbb{R}^3$ is a **smooth (embedded) surface** if for every $\mathbf{p} \in S$ there exists an open set $W \subset \mathbb{R}^3$, and open set $V \subset \mathbb{R}^2$ and a parametrization (or **surface patch**) $\boldsymbol{\sigma} : V \to S \subset \mathbb{R}^3$ so that:

- $\mathbf{p} \in U$ where $U = S \cap W$, and $\boldsymbol{\sigma} : V \to U \subset \mathbb{R}^3$ is a bijection from V to U,
- $\boldsymbol{\sigma}(u, v) = (x(u, v), y(u, v), z(u, v))$ is smooth (i.e. has partial derivatives of all orders),
- $\boldsymbol{\sigma}$ is regular: for all $(u, v) \in V$, the derivative $D_{(u,v)}\boldsymbol{\sigma} : \mathbb{R}^2 \to \mathbb{R}^3$ is injective.

Definition 13. Given a surface S, the **Tangent Space** of a point $\mathbf{p} \in S$, denoted by $T_{\mathbf{p}}S$, is the *SUBSET* in $\mathbb{R}^3$ spanned by $\sigma_u(\mathbf{p})$ and $\sigma_v(\mathbf{p})$ at point $\mathbf{p}$.

Definition 14. Let $\mathbf{p}$ be a point of a smooth surface $S \subset \mathbb{R}^3$. The **first fundamental form of** S **at** $\mathbf{p}$ is the inner product $\langle \cdot, \cdot \rangle_{\mathbf{p},S}$ on $T_{\mathbf{p}}S \subset \mathbb{R}^3$ defined by $\langle \mathbf{v}, \mathbf{w} \rangle_{\mathbf{p},S} = \mathbf{v} \cdot \mathbf{w}$for $\mathbf{v}, \mathbf{w} \in T_{\mathbf{p}}S$.

For a surface patch $\boldsymbol{\sigma} : V \to U \subset S$, the first fundamental form (in local coordinates) is the Riemannian metric on V defined by, for $\mathbf{q} \in V$ and $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$,$\langle \mathbf{v}, \mathbf{w} \rangle_{\mathbf{q}} = \langle D_{\mathbf{q}}\boldsymbol{\sigma}(\mathbf{v}), D_{\mathbf{q}}\boldsymbol{\sigma}(\mathbf{w}) \rangle_{\boldsymbol{\sigma}(\mathbf{q}),S} = D_{\mathbf{q}}\boldsymbol{\sigma}(\mathbf{v}) \cdot D_{\mathbf{q}}\boldsymbol{\sigma}(\mathbf{w})$

Definition 15. Let $\sigma : V \to S$ be a surface patch. The two-variable form of Taylor's theorem tells us that, near a point $\mathbf{p} = (u_0, v_0) \in V$,

$$\boldsymbol{\sigma}(u_0 + h, v_0 + k) - \boldsymbol{\sigma}(u_0, v_0) = \boldsymbol{\sigma}_u(\mathbf{p})\, h + \boldsymbol{\sigma}_v(\mathbf{p})\, k + \frac{1}{2}\left(\boldsymbol{\sigma}_{uu}(\mathbf{p})\, h^2 + 2\boldsymbol{\sigma}_{uv}(\mathbf{p})\, hk + \boldsymbol{\sigma}_{vv}(\mathbf{p})\, k^2\right) + o\left(h^2 + k^2\right).$$

Taking dot products with $\mathbf{N} = \mathbf{N}_{\boldsymbol{\sigma}}(\boldsymbol{\sigma}(u_0, v_0))$ we have

$$(\boldsymbol{\sigma}(u_0 + h, v_0 + k) - \boldsymbol{\sigma}(u_0, v_0)) \cdot \mathbf{N} = \frac{1}{2}\left(Lh^2 + 2Mhk + Nk^2\right) + o\left(h^2 + k^2\right)$$

The bilinear form at $T_{\mathbf{p}}S$ defined by: $\langle\langle \mathbf{v}, \mathbf{w} \rangle\rangle_{\mathbf{p},S} = \mathbf{v}^T \begin{pmatrix} L & M \\ M & N \end{pmatrix}\Big|_{\mathbf{p}} \mathbf{w}$ is called the **second fundamental form** of S at $\mathbf{p}$.

Definition 16. The **principal curvatures** κ_1, κ_2 **/ vectors at** $\mathbf{p}$ are the eigenvalues / eigenspace of the matrix $A|_{\mathbf{p}} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \begin{pmatrix} L & M \\ M & N \end{pmatrix}\Big|_{\mathbf{p}}$. The **mean curvature** is $H = \dfrac{\kappa_1 + \kappa_2}{2}$, the **Gaussian curvature** is $K = \kappa_1\kappa_2$.

Remark: If we fix a point $\mathbf{p} \in S$:

- There is always a one to one correspondence between the eigenvector of A considered in $\mathbb{R}^2$ and $T_{\mathbf{p}}S$: $(x, y) \to x\boldsymbol{\sigma}_u(\mathbf{p}) + y\boldsymbol{\sigma}_v(\mathbf{p})$
- The principal curvatures exists and are real, if they are distinct (i.e. spans $T_{\mathbf{p}}S$), then they are orthogonal.
- If $\mathbf{v}_1, \mathbf{v}_2$ is the corresponding eigenvector of A at $\mathbf{p}$, then $\langle\langle \mathbf{v}_i, \mathbf{v}_j \rangle\rangle_{\mathbf{p},S} = \langle \mathbf{v}_i, \kappa_j \mathbf{v}_j \rangle_{\mathbf{p},S} = \delta_{ij}$

Definition 17. Suppose $S, \widetilde{S} \subset \mathbb{R}^3$ are smooth surfaces and that $\mathbf{f} : \widetilde{S} \to S$ is a smooth function. We say that $\mathbf{f}$ is an **local isometry** if for any surface patches $\boldsymbol{\sigma} : V \to U \subset S$ and $\tilde{\boldsymbol{\sigma}} : \widetilde{V} \to \widetilde{U} \subset \widetilde{S}$ with $\mathbf{f}(U) = \widetilde{U}$, we have that $\boldsymbol{\sigma}^{-1} \circ \mathbf{f} \circ \tilde{\boldsymbol{\sigma}} : \widetilde{V} \to V$ is a Riemannian isometry with respect to the first fundamental forms $\langle \cdot, \cdot \rangle$ on U and $\langle \cdot, \cdot \rangle^{\sim}$ on $\widetilde{U}$. If $\mathbf{f}$ is, in addition, a diffeomorphism, it is called an **isometry**.

4 Curves on Surfaces, Gauss Bonnet

Theorem 10. *To find length of prove things above length in a surface S, always write $\boldsymbol{\gamma}(t) = \boldsymbol{\sigma}(u(t), v(t))$. Then theorem 2 and definition 5 agree. I.e. the length is*

$$\int_a^b \left(\langle \boldsymbol{\gamma}'(t), \boldsymbol{\gamma}'(t) \rangle_{\boldsymbol{\gamma}(t),S}\right)^{0.5} dt = \int_a^b \left(Eu'^2 + 2Fu'v' + Gv'^2\right)^{1/2} dt = \int_a^b \|\boldsymbol{\sigma}_u u' + \boldsymbol{\sigma}_v v'\|\, dt = \int_a^b \|\boldsymbol{\gamma}'\|\, dt$$

	Definition	Monge Patch $\boldsymbol{\sigma}(u,v) = (u, v, f(u,v))$	Surface of Revolution
E	$\boldsymbol{\sigma}_u \cdot \boldsymbol{\sigma}_u$	$1 + f_x^2$	$\dot{f}^2 + \dot{g}^2 = 1$
F	$\boldsymbol{\sigma}_u \cdot \boldsymbol{\sigma}_v$	$f_x f_y$	0
G	$\boldsymbol{\sigma}_u \cdot \boldsymbol{\sigma}_u$	$1 + f_y^2$	f^2
$\mathbf{N}$	$\dfrac{\boldsymbol{\sigma}_u \times \boldsymbol{\sigma}_v}{\|\boldsymbol{\sigma}_u \times \boldsymbol{\sigma}_v\|}$	$\left(1 + f_x^2 + f_y^2\right)^{-1/2} (-f_x, -f_y, 1)$	$(-\dot{g}\cos v, -\dot{g}\sin v, \dot{f})$
L	$\boldsymbol{\sigma}_{uu} \cdot \mathbf{N}$	$\left(1 + f_x^2 + f_y^2\right)^{-1/2} f_{xx}$	$\dot{f}\ddot{g} - \ddot{f}\dot{g}$
M	$\boldsymbol{\sigma}_{uv} \cdot \mathbf{N}$	$\left(1 + f_x^2 + f_y^2\right)^{-1/2} f_{xy}$	0
N	$\boldsymbol{\sigma}_{vv} \cdot \mathbf{N}$	$\left(1 + f_x^2 + f_y^2\right)^{-1/2} f_{yy}$	$f\dot{g}$
H	$\dfrac{GL - 2FM + EN}{EG - F^2}$	$\dfrac{\left(1 + f_y^2\right) f_{xx} - 2 f_x f_y f_{xy} + \left(1 + f_x^2\right) f_{yy}}{2\left(1 + f_x^2 + f_y^2\right)^{3/2}}$	0
K	$\dfrac{LN - M^2}{EG - F^2}$	$\dfrac{f_{xx} f_{yy} - f_{xy}^2}{\left(1 + f_x^2 + f_y^2\right)^2}$	$\dfrac{(\dot{f}\ddot{g} - \ddot{f}\dot{g})\dot{g}}{f} = -\dfrac{\ddot{f}}{f}$

Definition 18. Suppose $\boldsymbol{\gamma} : (a,b) \to S$ is a unit-speed smooth curve in a oriented smooth surface S. The **normal curvature** $\kappa_n = \kappa_n(t)$ and **geodesic curvature** $\kappa_g = \kappa_g(t)$ of $\boldsymbol{\gamma}$ at time $t \in (a,b)$ are defined by the following:

$$\boldsymbol{\gamma}''(t) = \kappa_n(t)\mathbf{N}(\boldsymbol{\gamma}(t)) + \kappa_g(t)\mathbf{N}(\boldsymbol{\gamma}(t)) \times \boldsymbol{\gamma}'(t)$$

This is well-defined because $\boldsymbol{\gamma}'(t)$ is (a unit vector) orthogonal to $\mathbf{N}$ thus $\boldsymbol{\gamma}'(t), \mathbf{N}(\boldsymbol{\gamma}(t))$ and $\mathbf{N}(\boldsymbol{\gamma}(t)) \times \boldsymbol{\gamma}'(t)$ are all linearly independent. Since $\boldsymbol{\gamma}''(t)$ is orthogonal to $\boldsymbol{\gamma}'(t)$ we can write it in a unique way as above.

Theorem 11. *If $\boldsymbol{\gamma} : (a,b) \to S$ is a smooth regular curve in an oriented smooth surface, then*

$$\kappa_n(t) = \frac{\langle\langle \boldsymbol{\gamma}'(t), \boldsymbol{\gamma}'(t)\rangle\rangle_{\boldsymbol{\gamma}(t),S}}{\langle \boldsymbol{\gamma}'(t), \boldsymbol{\gamma}'(t)\rangle_{\boldsymbol{\gamma}(t),S}} = \frac{Lu'(t)^2 + 2Mu'(t)v'(t) + Nv'(t)^2}{Eu'(t)^2 + 2Fu'(t)v'(t) + Gv'(t)^2}, \kappa_g = \frac{(\mathbf{N} \times \boldsymbol{\gamma}') \cdot \boldsymbol{\gamma}''}{\|\boldsymbol{\gamma}'\|^3}.$$

where L, M, N, E, F, G are functions / evaluated at $\boldsymbol{\gamma}(t)$.

Definition 19. Suppose $\boldsymbol{\sigma} : V \to U \subset S \subset \mathbb{R}^3$ is a surface patch. A **curvilinear polygon** in U is a continuous map $\boldsymbol{\gamma} : [0,T] \to U$ so that there exists $0 = t_0 < t_1 < \ldots < t_n = T$ with

- Simple Closed: $\boldsymbol{\gamma}(0) = \boldsymbol{\gamma}(T)$, and $\boldsymbol{\gamma}(t) \neq \boldsymbol{\gamma}(t')$ for all $0 \leq t < t' < T$;
- the restrictions $\boldsymbol{\gamma} : (t_i, t_{i+1}) \to U$ are all smooth regular curves;
- Enclosure: writing $\boldsymbol{\gamma}(t) = \boldsymbol{\sigma}(u(t), v(t))$, the curve $t \mapsto (u(t), v(t)) \in \mathbb{R}^2$ encloses a region D entirely in V, with image $\boldsymbol{\sigma}(D) \subset U$ called $\text{int}(\gamma) = \boldsymbol{\sigma}(D)$;
- for each i, $\boldsymbol{\gamma} : [t_{i-1}, t_i] \to U$ and $\boldsymbol{\gamma} : [t_i, t_{i+1}] \to U$ make an interior angle $\alpha_i \in (0, 2\pi)$ at $\boldsymbol{\gamma}(t_i)$.

If the region D is to the left of $(u(t), v(t))$ as t increases (anti-clockwise parametrization), we say that $\boldsymbol{\gamma}$ is positively oriented.

Theorem 12 (Gauss-Bonnet)**.** *for curvilinear polygons Suppose $\boldsymbol{\sigma} : V \to U \subset S \subset \mathbb{R}^3$ is a surface patch. Let $\gamma : [0,T] \to U$ be a positively oriented, unit-speed, curvilinear polygon in U, with interior angles $\alpha_1, \ldots, \alpha_n$ and enclosing a region int (γ) Then*

$$\int_\gamma \kappa_g ds = 2\pi - \sum_{i=1}^{n} (\pi - \alpha_i) - \int_{\text{int}(\gamma)} K dS$$

Definition 20. Let $S \subset \mathbb{R}^3$ be a closed, bounded, smooth surface. A (smooth) **triangulation** of S is a collection of curvilinear polygons in S so that

- each curvilinear polygon, together with the region it encloses, is contained in some surface patch;
- each point in S is in at least one of the curvilinear polygons;
- two curvilinear polygons are either disjoint, or their intersection is a common edge or a common vertex;
- each edge is an edge of exactly two polygons.

Theorem 13. *Every close bounded surface have a (finite) triangulation. From here we can count the **Euler Characteristics** $\chi = V - E + F$, and the total gaussian curvature of the surface $\int_S K\, dA = 2\pi\chi(S)$.*